

# DISCRIMINATIVE PROTEIN SEQUENCE MODELLING WITH LATENT SPACE DIFFUSION

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## ABSTRACT

We explore a framework for protein sequence representation learning that decomposes the task between manifold learning and distributional modelling. Specifically we present a Latent Space Diffusion architecture which combines a protein sequence autoencoder with a denoising diffusion model operating on its latent space. We obtain a one-parameter family of learned representations from the diffusion model, along with the autoencoder’s latent representation. We propose and evaluate two autoencoder architectures: a homogeneous model forcing amino acids of the same type to be identically distributed in the latent space, and an inhomogeneous model employing a noise-based variant of masking. As a baseline we take a latent space learned by masked language modelling, and evaluate discriminative capability on a range of protein property prediction tasks. Our finding is twofold: the diffusion models trained on both our proposed variants display higher discriminative power than the one trained on the masked language model baseline, none of the diffusion representations achieve the performance of the masked language model embeddings themselves.

## 1 INTRODUCTION

Proteins are an important class of biomolecules whose function, interaction, and evolutionary relationships are central to understanding cellular mechanisms and the complexity of life. While the underlying principles governing proteins and their behaviour admit explicit formulations in quantum chemistry, these are in practice too complex to model directly. The quest for simplifying representations and approximations which strike a balance between generality, accuracy and computational efficiency is a core challenge of computational biology.

Machine learning provides a powerful suite of tools for representation learning. A prominent method is an autoencoder employing an information bottleneck to learn a compressed latent representation Tishby et al. (2000). For proteins this is less applicable however, as a protein’s primary sequence already provides an incredibly compact representation (each amino acid, being a categorical variable of size 20, can be represented with just 5 bits). Indeed the primary sequence completely determines a protein, and the key challenge is how to decode from this. Applications of machine learning to protein representation learning from sequence data can be roughly organised around two main threads: those such as AlphaFold which leverage multiple sequence alignments (MSAs) capturing co-evolutionary information Jumper et al. (2021); Rao et al. (2021); Truong Jr & Bepler (2023), and those such as ESM Rives et al. (2021); Lin et al. (2023); Hayes et al. (2024) utilising masked language modelling (MLM) Devlin et al. (2019); Elnaggar et al. (2021); Brandes et al. (2022).

Generative modelling is closely tied to representation learning Kingma & Welling (2014); Goodfellow et al. (2014). Indeed masked language modelling is a form of reconstructive learning, where a model is trained to restore partially corrupted input, which underlies its ability to learn rich contextually aware representations Devlin et al. (2019). For continuous spaces, Gaussian diffusion has emerged as a leading generative method due to its ability to produce diverse high-quality samples from complex distributions Sohl-Dickstein et al. (2015); Ho et al. (2020); Song et al. (2020). By learning how Gaussian noise diffuses through a data space, a diffusion model learns to approximate the score function of the data distribution,  $s(x) = \nabla_x \log p_{data}(x)$ . From a statistical physics perspective, expressing the distribution in Boltzmann form  $p_{data}(x) \sim \exp(-E(x))$ , the score function

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admits a natural interpretation as a distributional force:  $F(x) \equiv -\nabla_x E(x) = s(x)$ , which underlies a diffusion model’s ability to both navigate the distribution efficiently as well as to learn a meaningful representation of the data Song et al. (2020); Vincent et al. (2008).

Studies of generative modelling on protein sequence data have primarily focused on discrete diffusion methods Gruver et al. (2024); Alamdari et al. (2023); Wang et al. (2024). These are counterparts to Gaussian diffusion that place emphasis on the categorical nature of amino acids. In contrast to the continuous setting however, the formalism of discrete diffusion is less well established, and principled approaches have appeared only relatively recently Campbell et al. (2024); Shi et al. (2024); Zheng et al. (2024); Gat et al. (2024). Another notable work is DPLM Wang et al. (2024), an adaption of masked language modelling, where in particular the authors highlight the representation learning capabilities of their generative model and demonstrate it achieves competitive performance with ESM2.

Let us question however whether a masking-based approach is the best route towards modelling protein sequence data. From a reconstructive learning perspective, it is unclear whether masking is an optimal way to represent a corrupted sequence. For instance, the unmasked amino acids are fully specified with no ambiguity, while for the masked amino acids only their ambiguity is conveyed. One can imagine instead an alternative corruption process where partial information is retained/erased, for example expected physiochemical properties or aspects of long-range dependencies. Similarly from a generative perspective, one may question the task of performing distributional modelling directly in the discrete domain, as at the level of sequence the protein landscape is far from smooth: single mutations can have abrupt consequences while compound mutations may be strongly correlated.

This motivates us to explore a switch in focus from a discrete representation of sequence space to a continuous one. This can be framed as making a distinction between two aspects of protein representation learning: manifold learning and distribution learning. The first addresses the question of how to embed protein sequences in a continuous latent space, while the second concerns the distribution of protein sequences over this latent space. Here Gaussian diffusion can be employed for the distributional modelling, and so the question then is how to learn an appropriate latent space.

Previous works adopting a latent diffusion approach for protein sequence data examined the use of the ESM embeddings Meshchaninov et al. (2024); Chen et al. (2024); Lu et al. (2024a). There is a difficulty with this approach however, which can be attributed to the embeddings retaining much of the discreteness of the underlying sequence Li et al. (2023). In essence, amino acid embeddings are too robust to added noise, which obstructs the learning ability of denoising. Indeed a parallel can be made here to high-resolution images, for which latent diffusion models were originally introduced Rombach et al. (2022).

In this work we attempt to address the challenge of how to construct a latent space which facilitates the distributional modelling of proteins sequence data. To this end we propose two novel sequence autoencoder architectures: a homogeneous model forcing amino acids of the same type to be identically distributed in the latent space, and an inhomogeneous model employing a noise-based variant of masking. We train a diffusion model on their latent space, and identify how this gives rise to an additional one-parameter family of learned representations. We focus on this discriminative capability of the diffusion model, and evaluate it on a diverse set of representation learning benchmarks.

## 2 RELATED WORK

The use of diffusion/denoising for protein representation learning was introduced for the structural representation Zaidi et al. (2022); Liu et al. (2022), based on a connection between the learned score function and molecular force fields. DSMBind employs SE(3) denoising score matching as an unsupervised pre-training task for binding energy prediction Jin et al. (2023). From a generative perspective, diffusion has also been mostly applied to structure Watson et al. (2023); Ingraham et al. (2023); Yim et al. (2023); Lee et al. (2023), for which the Gaussian form of diffusion can be employed. On protein structure prediction, AlphaFold 3 Abramson et al. (2024) trains a conditioned diffusion model for the generation of its structural predictions, and PLAID Lu et al. (2024a) takes a latent diffusion approach. LatentDock McPartlon et al. (2023) employs structure-based latent

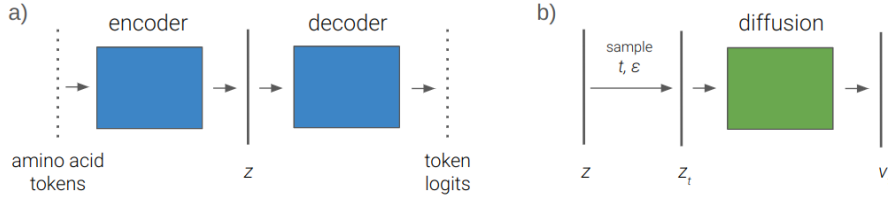


Figure 1: The LSD model is comprised of **(a)** a protein sequence autoencoder which learns a latent space  $z$ , and **(b)** a diffusion model acting on this latent space. The autoencoder is trained end-to-end by balancing a reconstruction loss, between input amino acid tokens and output token logits, against a normalization loss on the distribution of latent space embeddings. We consider two variants, LSD-TN with a non-trivial normalization loss and LSD-NM with a non-trivial reconstruction loss, as described in Sec. 3. The diffusion model learns to map noised latent embeddings  $z_t = \cos(\pi t/2)z + \sin(\pi t/2)\varepsilon$  to their orthogonal complements  $v_t = -\sin(\pi t/2)z + \cos(\pi t/2)\varepsilon$ , which thereby provides an additional one-parameter family of sequence representation to that obtained at the latent space.

diffusion for modelling protein-protein docking. Application to the generation of conformational ensembles has also been explored Jing et al. (2024); Hassan et al. (2024).

Discrete diffusion applied on protein sequence data has been explored in LaMBO-2 Gruver et al. (2024), EvoDiff Alamdari et al. (2023), and DPLM Wang et al. (2024). DPLM stands out for evaluating the representation learning capabilities of their model, demonstrating it’s ability to perform competitively across a range of prediction tasks. There are also works exploring latent diffusion on language model embeddings. DiMA Meshchaninov et al. (2024) and AMP-Diffusion (focused solely on antimicrobial peptides) Chen et al. (2024) employed pre-trained ESM2 embeddings, and PLAID Lu et al. (2024a) employed the compressed CHEAP Lu et al. (2024b) embeddings of ESM2. Both DiMA and PLAID trained their diffusion models with the heuristic method of self-conditioning Chen et al. (2022).

Two studies explored latent diffusion on pre-trained ESM2 embeddings Chen et al. (2024); Meshchaninov et al. (2024), and see also PLAID Lu et al. (2024a) which employs the compressed CHEAP Lu et al. (2024b) embeddings of ESM.

More broadly, the link between discriminative and generative modelling underlies the auto-regressive approach Ferruz et al. (2022); Madani et al. (2023), the masked language modelling approach Rives et al. (2021); Lin et al. (2023); Hayes et al. (2024); Elnaggar et al. (2021); Brandes et al. (2022), as well as variational autoencoder approaches Sinai et al. (2017); Sevgen et al. (2023), to modelling protein sequence data.

Beyond the field of protein modelling, denoising autoencoders date back to the seminal work Vincent et al. (2008). Diffusion-based representation learning was advanced in Abstreiter et al. (2021). Latent diffusion models were introduced in the image domain Rombach et al. (2022), and the application of latent diffusion to discrete data has been predominantly studied in the natural language processing literature Li et al. (2022); Dieleman et al. (2022); Strudel et al. (2022); Gao et al. (2024); Ye et al. (2024); Gulrajani & Hashimoto (2024).

### 3 LATENT SPACE DIFFUSION

We employ a Latent Space Diffusion (LSD) architecture, illustrated in Fig. 1, comprised of

- an autoencoder learning a latent manifold embedding of protein sequences,
- a diffusion model for the distributional modelling of protein sequence datasets over this learned latent space.

We adopt a transformer-based architecture for each component as shown in Appendix A. The autoencoder is composed of an encoder-decoder pair, which we set to have an equal number of layers. The encoder takes as input tokenized protein sequences, and outputs latent embeddings  $z_{a,i}$